Unshifted Moments of TBF Pair Distribution Function

As always the 1D complex Gaussian wavepackets (CGWPs) of AIMS are of the form

$$\chi_i(x; q_i, p_i, a_i) = \langle x | \, q_i, p_i, a_i \rangle = \left(\frac{2a}{\pi}\right)^{1/4} \exp\left(-a(x-q)^2 + ip(x-q)\right) , \tag{1}$$

The overlap between two such Gaussians χ_i and χ_j is

$$S_{ij} \equiv \langle \chi_i | \chi_j \rangle = \sqrt{\frac{2\sqrt{a_i a_j}}{a_{ij}}} \exp\left[-\frac{a_i a_j}{a_{ij}} \Delta q_{ij}^2 + i\omega_{ij} - \frac{\Delta p_{ji}^2}{4a_{ij}} + i q_{ij} \Delta p_{ji}\right] \tag{2}$$

To condense the notation slightly we have defined the following variables

$$\Delta p_{ji} = p_j - p_i \tag{3}$$
$$\Delta q_{ij} = q_i - q_j \tag{4}$$
$$a_{ij} = a_i + a_j \tag{5}$$
$$q_{ij} = \frac{a_i q_i + a_j q_j}{a_i + a_j} \tag{6}$$
$$\omega_{ij} = q_i p_i - q_j p_j \, . \tag{7}$$

As shall become clearer later we can rewrite the overlap expression ever so slightly

$$S_{ij} = \sqrt{\frac{2\sqrt{a_i a_j}}{a_{ij}}} \exp\left[-\frac{a_i a_j}{a_{ij}} \Delta q_{ij}^2 + i(q_i p_i - q_j p_j) + \frac{(i\Delta p_{ji})^2}{4a_{ij}} + i q_{ij} \Delta p_{ji}\right] . \tag{8}$$

Here we shall focus on the unshifted moments of the (complex) pair distribution function $\chi_i^* \chi_j$, that is, the integrals

$$m_{ij,n} \equiv \langle \chi_i | x^n | \chi_j \rangle = \int dx \, x^n \chi_i^*(x) \chi_j(x) \, . \tag{9}$$

Plugging the definition of the CGWPs into this expression one can see that

$$\begin{aligned}
m_{ij,n} &= \sqrt{\frac{2\sqrt{a_i a_j}}{\pi}} \int dx \, \exp\left(-\frac{a_i a_j}{a_{ij}} \Delta q_{ij} + i\omega_{ij} - a_{ij}(x - q_{ij})^2 + i\Delta p_{ji} x\right) x^n \\
&= \sqrt{\frac{2\sqrt{a_i a_j}}{\pi}} \int dx \, \frac{\partial^n}{\partial (i\Delta p_{ji})^n} \exp\left(-\frac{a_i a_j}{a_{ij}} \Delta q_{ij} + i\omega_{ij} - a_{ij}(x - q_{ij})^2 + i\Delta p_{ji} x\right) \\
&= \frac{\partial^n}{\partial (i\Delta p_{ji})^n} \left[\sqrt{\frac{2\sqrt{a_i a_j}}{\pi}} \int dx \, \exp\left(-\frac{a_i a_j}{a_{ij}} \Delta q_{ij} + i\omega_{ij} - a_{ij}(x - q_{ij})^2 + i\Delta p_{ji} x\right)\right] \\
&= \frac{\partial^n}{\partial (i\Delta p_{ji})^n} S_{ij}
\end{aligned} \tag{10}$$

The derivatives are straight forward to calculate, as we shall now do

$$m_{ij,n} = \frac{\partial^n}{\partial (i\Delta p_{ji})^n} S_{ij} = \frac{\partial^{n-1}}{\partial (i\Delta p_{ji})^{n-1}} \left[\left(\frac{i\Delta p_{ji}}{2a_{ij}} + q_{ij}\right) S_{ij}\right] . \tag{11}$$

Denoting $z_{ij} = \frac{i\Delta p_{ji}}{2a_{ij}} + q_{ij}$, it is clear then that $m_{ij,1} = z_{ij}S_{ij}$, which is the first moment. Using the chain-rule we get

$$\begin{aligned} m_{ij,n} &= \frac{\partial^{n-1}}{\partial(i\Delta p_{ji})^{n-1}}\left[z_{ij}S_{ij}\right] \\ &= \frac{\partial^{n-2}}{\partial(i\Delta p_{ji})^{n-2}}\left[S_{ij}\frac{\partial}{\partial(i\Delta p_{ji})}z_{ij} + z_{ij}\frac{\partial}{\partial(i\Delta p_{ji})}S_{ij}\right] \\ &= \frac{\partial^{n-2}}{\partial(i\Delta p_{ji})^{n-2}}\left[S_{ij}\frac{1}{2a_{ij}} + z_{ij}^2 S_{ij}\right] \\ &= \frac{\partial^{n-2}}{\partial(i\Delta p_{ji})^{n-2}}\left[\left(z_{ij}^2 + \frac{1}{2a_{ij}}\right)S_{ij}\right] \end{aligned} \tag{12}$$

Thus $m_{ij,2} = \left(z_{ij}^2 + \frac{1}{2a_{ij}}\right)S_{ij}$. Going to the next order we get

$$\begin{aligned} m_{ij,n} &= \frac{\partial^{n-2}}{\partial(i\Delta p_{ji})^{n-2}}\left[\left(z_{ij}^2 + \frac{1}{2a_{ij}}\right)S_{ij}\right] \\ &= \frac{\partial^{n-3}}{\partial(i\Delta p_{ji})^{n-3}}\left[2z_{ij}\frac{1}{2a_{ij}}S_{ij} + \left(z_{ij}^2 + \frac{1}{2a_{ij}}\right)z_{ij}S_{ij}\right] \\ &= \frac{\partial^{n-3}}{\partial(i\Delta p_{ji})^{n-3}}\left[\left(z_{ij}^3 + 3z_{ij}\frac{1}{2a_{ij}}\right)S_{ij}\right], \end{aligned} \tag{13}$$

that is, $m_{ij,3} = \left(z_{ij}^3 + 3z_{ij}\frac{1}{2a_{ij}}\right)S_{ij}$.

$$\begin{aligned} m_{ij,n} &= \frac{\partial^{n-3}}{\partial(i\Delta p_{ji})^{n-3}}\left[\left(z_{ij}^3 + 3z_{ij}\frac{1}{2a_{ij}}\right)S_{ij}\right] \\ &= \frac{\partial^{n-4}}{\partial(i\Delta p_{ji})^{n-4}}\left[\left(3z_{ij}^2\frac{1}{2a_{ij}} + 3\frac{1}{(2a_{ij})^2}\right)S_{ij} + \left(z_{ij}^3 + 3z_{ij}\frac{1}{2a_{ij}}\right)z_{ij}S_{ij}\right] \\ &= \frac{\partial^{n-4}}{\partial(i\Delta p_{ji})^{n-4}}\left[\left(z_{ij}^4 + 6z_{ij}^2\frac{1}{2a_{ij}} + 3\frac{1}{(2a_{ij})^2}\right)S_{ij}\right], \end{aligned} \tag{14}$$

so $m_{ij,4} = \left(z_{ij}^4 + 6z_{ij}^2\frac{1}{2a_{ij}} + 3\frac{1}{(2a_{ij})^2}\right)S_{ij}$.

$$\begin{aligned} m_{ij,n} &= \frac{\partial^{n-4}}{\partial(i\Delta p_{ji})^{n-4}}\left[\left(z_{ij}^4 + 6z_{ij}^2\frac{1}{2a_{ij}} + 3\frac{1}{(2a_{ij})^2}\right)S_{ij}\right] \\ &= \frac{\partial^{n-5}}{\partial(i\Delta p_{ji})^{n-5}}\left[\left(4z_{ij}^3\frac{1}{2a_{ij}} + 12z_{ij}\frac{1}{(2a_{ij})^2}\right)S_{ij} + \left(z_{ij}^4 + 6z_{ij}^2\frac{1}{2a_{ij}} + 3\frac{1}{(2a_{ij})^2}\right)z_{ij}S_{ij}\right] \\ &= \frac{\partial^{n-5}}{\partial(i\Delta p_{ji})^{n-5}}\left[\left(z_{ij}^5 + 10z_{ij}^3\frac{1}{2a_{ij}} + 15z_{ij}\frac{1}{(2a_{ij})^2}\right)S_{ij}\right], \end{aligned} \tag{15}$$

thus $m_{ij,5} = \left(z_{ij}^5 + 10z_{ij}^3\frac{1}{2a_{ij}} + 15z_{ij}\frac{1}{(2a_{ij})^2}\right)S_{ij}$. The next order should make clear the pattern that is emerging from these derivatives:

$$\begin{aligned} m_{ij,n} &= \frac{\partial^{n-4}}{\partial(i\Delta p_{ji})^{n-4}}\left[\left(z_{ij}^5 + 10z_{ij}^3\frac{1}{2a_{ij}} + 15z_{ij}\frac{1}{(2a_{ij})^2}\right)S_{ij}\right] \\ &= \frac{\partial^{n-6}}{\partial(i\Delta p_{ji})^{n-6}}\left[\left(5z_{ij}^4\frac{1}{2a_{ij}} + 30z_{ij}^2\frac{1}{(2a_{ij})^2} + 15\frac{1}{(2a_{ij})^3}\right)S_{ij} + \left(z_{ij}^5 + 10z_{ij}^3\frac{1}{2a_{ij}} + 15z_{ij}\frac{1}{(2a_{ij})^2}\right)z_{ij}S_{ij}\right] \\ &= \frac{\partial^{n-6}}{\partial(i\Delta p_{ji})^{n-6}}\left[\left(z_{ij}^6 + 15z_{ij}^4\frac{1}{2a_{ij}} + 45z_{ij}^2\frac{1}{(2a_{ij})^2} + 15\frac{1}{(2a_{ij})^3}\right)S_{ij}\right], \end{aligned} \tag{16}$$

hence $m_{ij,6} = \left(z_{ij}^6 + 15z_{ij}^4 \frac{1}{2a_{ij}} + 45z_{ij}^2 \frac{1}{(2a_{ij})^2} + 15\frac{1}{(2a_{ij})^3}\right) S_{ij}$. It is not hard to see that

$$m_{ij,n} = S_{ij} \sum_{k=0}^{\lfloor n/2 \rfloor} T(n,k) z_{ij}^{n-2k} \frac{1}{(2a_{ij})^k} \tag{17}$$

where the coefficients $T(n,k)$ are the so-called Bessel numbers. To find some kind of pattern in them we can look at a table of the coefficients of the n-th moments up to $n = 9$. If no number is written, $T(n,k) = 0$.

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
$n=1$	1					
$n=2$	1	1				
$n=3$	1	3				
$n=4$	1	6	3			
$n=5$	1	10	15			
$n=6$	1	15	45	15		
$n=7$	1	21	105	105		
$n=8$	1	28	210	420	105	
$n=8$	1	36	378	1260	945	
$n=9$	1	45	630	3150	4725	945
$n=9$	1	55	990	6930	17325	10395

A couple of things should immediately catch ones eye: at even n the number of nonzero coefficients increases by one with respect to the previous odd order. Furthermore, we can see that

$$T(n,k) = T(n-1,k-1)(n-1-2(k-1)) + T(n-1,k) \quad \forall n > 1, k \in \{1,2,\ldots,\lfloor n/2 \rfloor\} \tag{18}$$

and that $\forall n > 1 : T(n,0) = T(n-1,0) = 1$. To convince oneself of this recursion it is good to look at some examples: $\mathrm{LHS} = T(5,2) = 15$, $\mathrm{RHS} = T(4,1)(4-2) + T(4,2) = 6 \cdot 2 + 3 = 15$; $\mathrm{LHS} = T(6,2) = 45$, $\mathrm{RHS} = T(5,1)(6-2) + T(5,2) = 10 \cdot 3 + 15 = 45$; etc. We shall now prove the validity of Eq (17) for all orders of n by induction. To be precise, since we have already looked at the base cases, we need only to show that at any n given

$$m_{ij,n} = S_{ij} \sum_{k=0}^{\lfloor n/2 \rfloor} T(n,k) z_{ij}^{n-2k} \frac{1}{(2a_{ij})^k} \tag{19}$$

that at $n+1$

$$m_{ij,n+1} = S_{ij} \sum_{k=0}^{\lfloor n/2 \rfloor} T(n+1,k) z_{ij}^{n+1-2k} \frac{1}{(2a_{ij})^k} \tag{20}$$

with $\forall k \in \{1,2,\ldots,\lfloor n/2 \rfloor\} : T(n+1,k) = T(n,k-1)(n-1-2(k-1)) + T(n,k)$ and $T(n+1,0) = T(n,0)$. We already know that

$$m_{ij,n+1} = \frac{\partial}{\partial(i\Delta p_{ji})} m_{ij,n}\,, \tag{21}$$

holds so we can simply plug in the expression for the n-th moment (17) to get

$$\begin{aligned} m_{ij,n+1} &= \frac{\partial}{\partial(i\Delta p_{ji})} S_{ij} \sum_{k=0}^{\lfloor n/2 \rfloor} T(n,k) z_{ij}^{n-2k} \frac{1}{(2a_{ij})^k} + S_{ij} \sum_{k=0}^{\lfloor n/2 \rfloor} T(n,k) \frac{\partial}{\partial(i\Delta p_{ji})} z_{ij}^{n-2k} \frac{1}{(2a_{ij})^k} \\ &= z_{ij} S_{ij} \sum_{k=0}^{\lfloor n/2 \rfloor} T(n,k) z_{ij}^{n-2k} \frac{1}{(2a_{ij})^k} + S_{ij} \sum_{k=0}^{\lfloor n/2 \rfloor} T(n,k)(n-2k) z_{ij}^{n-1-2k} \frac{1}{(2a_{ij})^{k+1}} \end{aligned}$$

which after some index reshuffling becomes

$$\begin{aligned}
m_{ij,n+1} &= S_{ij} \sum_{k=0}^{\lfloor n/2 \rfloor} T(n,k) z_{ij}^{n+1-2k} \frac{1}{(2a_{ij})^k} + S_{ij} \sum_{k=0}^{\lfloor n/2 \rfloor} T(n,k)(n-2k) z_{ij}^{n-1-2k} \frac{1}{(2a_{ij})^{k+1}} \\
&= S_{ij} \sum_{k=0}^{\lfloor n/2 \rfloor} T(n,k) z_{ij}^{n+1-2k} \frac{1}{(2a_{ij})^k} + S_{ij} \sum_{k=1}^{\lfloor n/2 \rfloor} T(n,k-1)(n-2(k-1)) z_{ij}^{n-1-2(k-1)} \frac{1}{(2a_{ij})^k} \\
&= S_{ij} \sum_{k=0}^{\lfloor n/2 \rfloor} T(n,k) z_{ij}^{n+1-2k} \frac{1}{(2a_{ij})^k} + S_{ij} \sum_{k=1}^{\lfloor n/2 \rfloor} T(n,k-1)(n-2(k-1)) z_{ij}^{n+1-2k} \frac{1}{(2a_{ij})^k} \\
&= S_{ij} \sum_{k=0}^{\lfloor n/2 \rfloor} T(n+1,k) z_{ij}^{n+1-2k} \frac{1}{(2a_{ij})^k} \qquad (22)
\end{aligned}$$

with $T(n+1,0) = T(n,0)$ and $\forall k \in \{1,2,\ldots,\lfloor n/2 \rfloor\} : T(n+1,k) = T(n,k-1)(n-2(k-1)) + T(n,k)$ which is exactly what we wanted to show.